

Data Visualization Project

- Team Technerds

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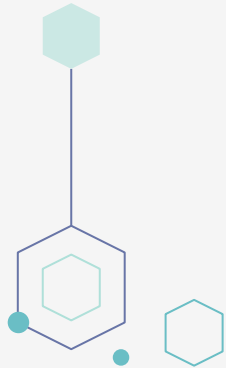


Overview

Project Overview : This project aims to analyze and visually explore factors influencing loan repayment behavior for a finance company operating in urban regions. The goal is to help the company better identify trustworthy applicants, reduce defaults, and strengthen the credit evaluation process.

Data: Analyzed **307,000+** current loan applications and **1.4 million+** historical records.

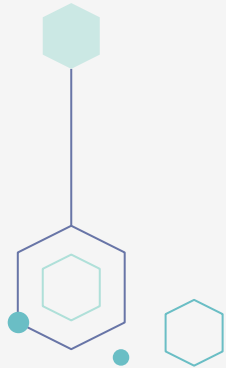
Key Objectives:

1. Identify defining characteristics of trustworthy vs. risky applicants.
 2. Analyze the impact of financial stability (Income, Annuity) on default risk.
 3. Develop an interactive dashboard to empower the credit team with real-time insights.
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Methodology & Data Strategy

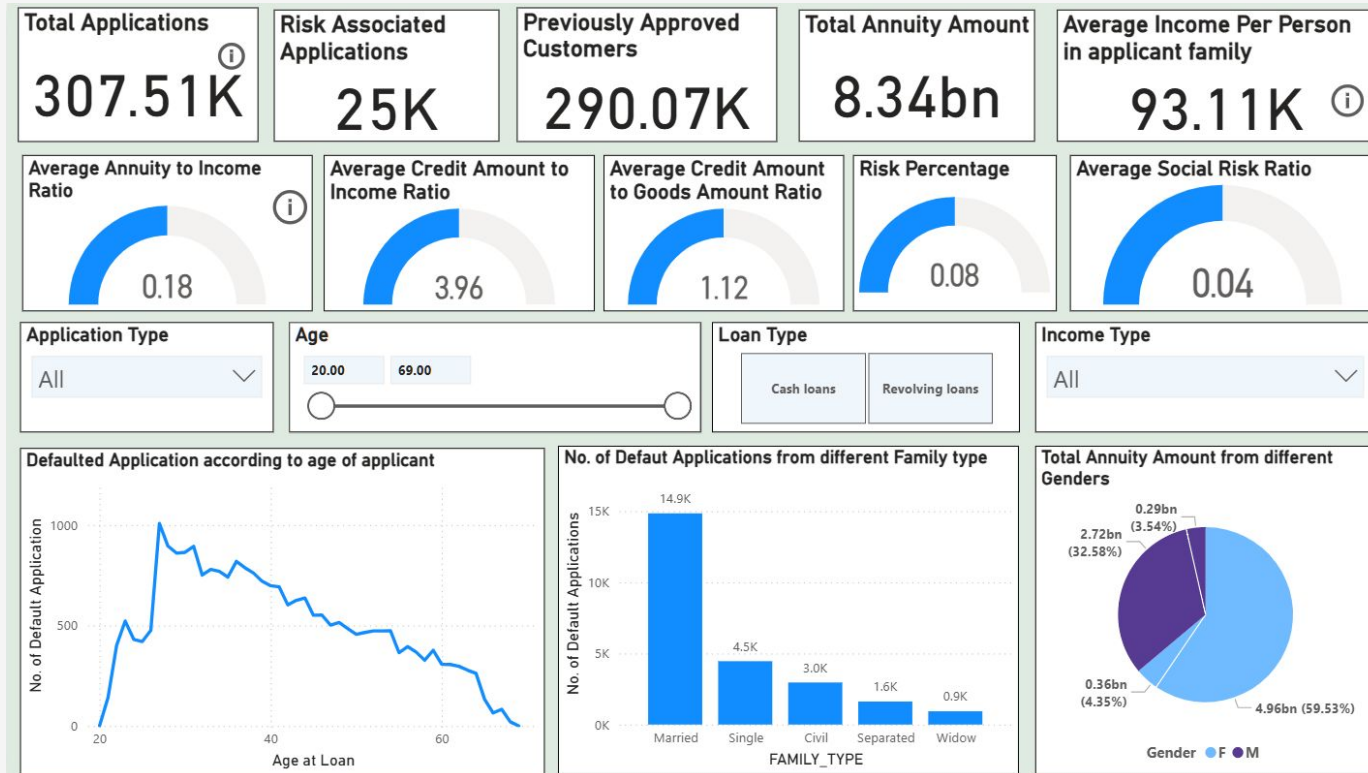
- **Data Inspection & Cleaning** : Handled missing values, treated outliers, and rectified data anomalies (e.g., employment errors).
- **Initial EDA** : Analyzed baseline distributions and identified the significant class imbalance (8% default rate).
- **Feature Engineering** : Developed domain-specific financial ratios (e.g., *Affordability*) and aggregated historical loan data.
- **Deep Data Analysis:** Conducted multivariate analysis, correlation heatmaps, and validated drivers via Feature Importance(via Random Forest with 0.75 roc score).
- **Power BI Dashboard** : Final interactive dashboard allowing stakeholders to segment risk and monitor portfolio health.

Dashboard



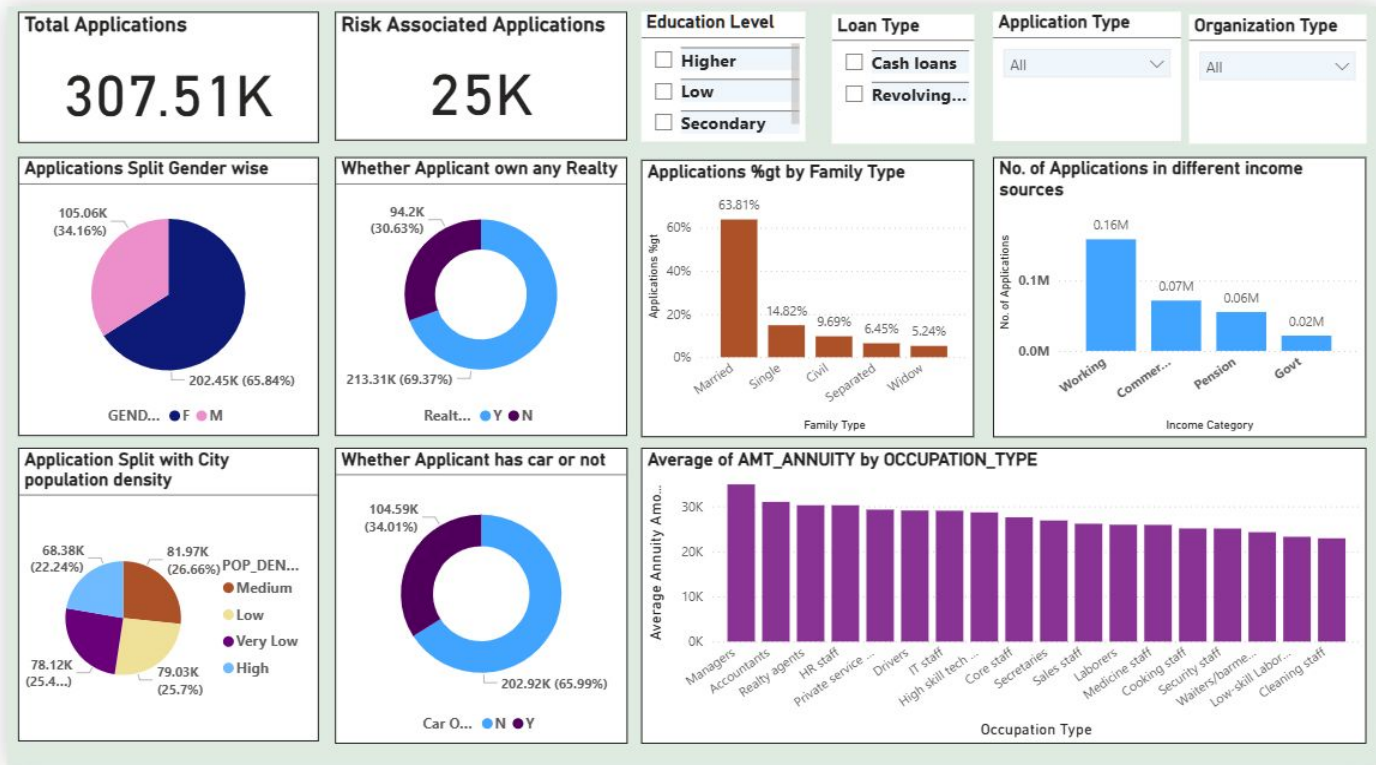
Overview Page

This page presents a consolidated view of loan applications, risk indicators, and key financial ratios, enabling quick assessment of portfolio health. Interactive filters allow users to drill down by applicant demographics, loan type, and income category to identify default trends and high-risk segments.



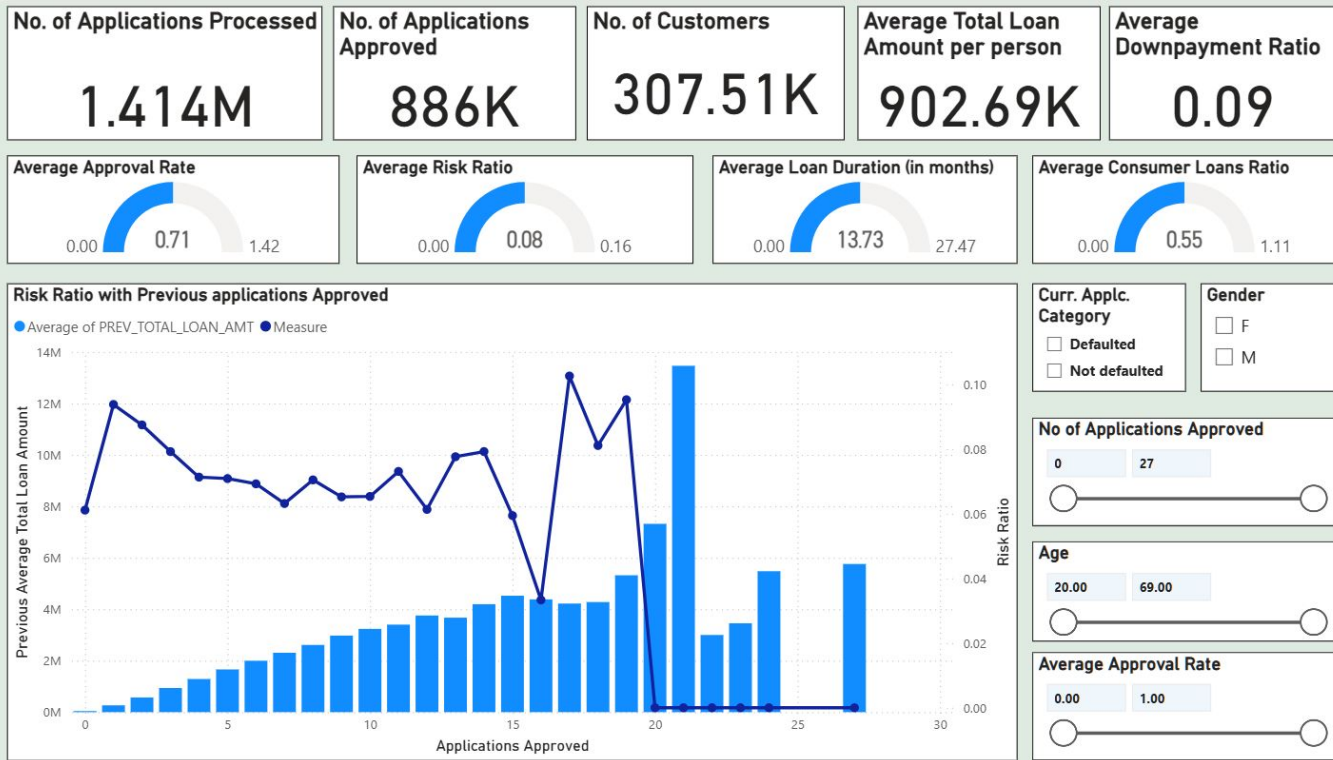
Current Applications Distribution

This page analyzes the distribution of loan applications across applicant demographics, asset ownership, family structure, income sources, and geographic density. It helps understand the composition of current applicants and identify dominant segments based on gender, education, occupation, and financial characteristics.



Previous Applications Distribution

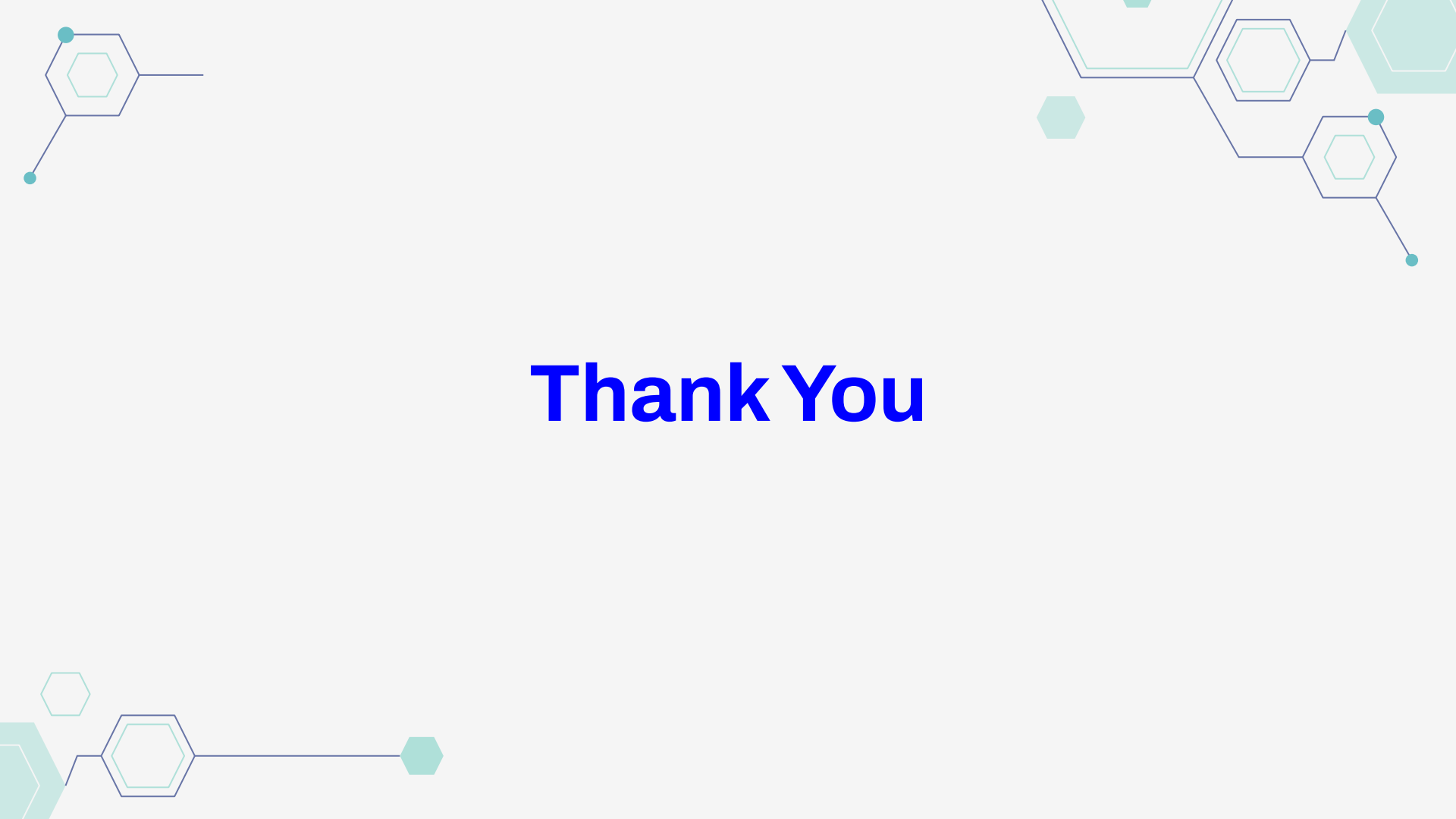
This page summarizes insights from already processed previous loan applications, highlighting approval volumes, approval rate, risk ratio, loan duration, and average loan characteristics. It enables analysis of historical approval behavior and associated risk patterns, with filters to examine trends across applicant demographics and approval outcomes.



Risk Drivers & Sensitivity Analysis Dashboard

This page focuses on analyzing the key parameters that strongly influence loan risk by using interactive gauges and high-correlation metrics such as income ratios, credit exposure, and annuity burden. Users can apply multiple slicer combinations to study how changes in applicant characteristics impact the risk ratio and default behavior, enabling deep sensitivity and what-if analysis of applicant risk profiles.



The slide features decorative chemical structures in the corners. The top-left corner shows a benzene ring with a teal dot at the top and a line extending to the right. The top-right corner shows a complex polycyclic structure with multiple benzene rings and teal dots. The bottom-left corner shows a benzene ring with a teal dot at the top and a line extending to the right. The bottom-right corner shows a benzene ring with a teal dot at the top and a line extending to the right.

Thank You